

[Chromium: Fact Sheet for Consumers \(original version\)](#).

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What is chromium and what does it do?

Chromium is a mineral found in many foods. Researchers do not fully understand what chromium does in your body, but it might help you use carbohydrates, fats, and proteins.

There are two main forms of chromium. This fact sheet is about chromium (III), the type of chromium found in foods and some dietary supplements. A different type of chromium, called chromium (VI) (sometimes called hexavalent chromium) is a poisonous byproduct of industrial manufacturing.

How much chromium do I need?

Scientists do not currently think that chromium is necessary for good health, and chromium deficiency has not been reported in healthy people. However, in 2001 scientists did consider chromium to be an essential nutrient, and they set recommended amounts based on the evidence available at that time. Here are the average daily recommended amounts in micrograms (mcg) that were established in 2001:

Life Stage	Recommended Amount
Birth to 6 months	0.2 mcg
Infants 7–12 months	5.5 mcg
Children 1–3 years	11 mcg
Children 4–8 years	15 mcg
Boys 9–13 years	25 mcg
Girls 9–13 years	21 mcg
Teen boys 14–18 years	35 mcg
Teen girls 14–18 years	24 mcg
Adult men 19–50 years	35 mcg
Adult women 19–50 years	25 mcg
Adult men 51+ years	30 mcg
Adult women 51+ years	20 mcg
Pregnant teens	29 mcg
Pregnant women	30 mcg

Life Stage	Recommended Amount
Breastfeeding teens	44 mcg
Breastfeeding women	45 mcg

What foods provide chromium?

Many foods contain chromium. The amount of chromium in fruits and vegetables depends on the amount of chromium in the soil and water in which they were grown. The amount in meat depends on the animal's diet. Using stainless steel equipment to process or cook foods can increase the chromium content of food because small amounts of chromium transfer to the food.

You can get recommended amounts of chromium by eating a variety of foods, including the following:

- Meats, such as ham, beef, and turkey
- Bread and other grain products
- Vegetables, such as lettuce and green beans
- Fruit, such as apples and bananas
- Juices, such as grape, orange, and tomato juices
- Brewer's yeast and nuts

What kinds of chromium dietary supplements are available?

Chromium is available in many dietary supplements, such as multivitamin/mineral supplements and supplements that contain only chromium. Chromium in dietary supplements is in many forms, including chromium picolinate and chromium chloride. Your body absorbs chromium similarly from the different forms used in supplements.

What are some effects of chromium supplements on health?

Scientists are studying chromium supplements to understand whether they affect health. Here are some examples of what this research has shown.

High blood sugar levels and diabetes

Chromium supplements are often promoted to improve blood sugar control in people with type 2 diabetes. However, studies examining the effects of chromium on high blood sugar levels or diabetes have had mixed results. The American Diabetes Association does not recommend chromium supplements for people with diabetes because these supplements don't have a clear benefit. More research is needed to understand whether chromium supplements might improve blood sugar control in some people.

Metabolic syndrome

Metabolic syndrome—the combination of excess belly fat, high levels of fat in the blood, high blood sugar, high blood pressure, and low levels of HDL (good) cholesterol—raises the risk of heart disease, diabetes, and stroke. A few clinical trials have studied the effect of chromium supplements on metabolic syndrome. These studies did not show a benefit of chromium supplementation in people with metabolic syndrome.

Polycystic ovary syndrome

Polycystic ovary syndrome (PCOS) is a hormonal disorder that affects females of reproductive age. Symptoms include irregular periods, infertility, excess hair on the face or body, acne, and weight gain. PCOS raises the risk of diabetes and heart disease. Studies examining the effects of chromium supplements on

PCOS have had mixed results. Any potential benefits appear to be very small. More research is needed to understand whether taking chromium might help lower the risk of PCOS or improve symptoms.

High cholesterol levels

High levels of LDL (bad) cholesterol can raise your risk of heart attack and stroke. Studies have examined whether chromium supplements improve cholesterol levels. The results of these studies have been mixed. More research is needed to understand whether chromium supplements have any effect on cholesterol levels.

Weight and lean body mass

Some chromium supplements are marketed for weight loss and to reduce body fat and increase muscle mass. However, clinical trials have found only a very small benefit. This benefit is unlikely to make any difference in health or appearance.

Can chromium be harmful?

Chromium from food and dietary supplements does not appear to cause any harm, but research is limited. People with kidney disease or liver disease should be cautious about taking high amounts of chromium.

Does chromium interact with medications or other dietary supplements?

Yes. Chromium supplements might interact or interfere with medicines that you take. Here are a few examples:

- Insulin. Taking chromium together with insulin might cause low blood sugar levels.
- Metformin and other antidiabetes medications. If you take antidiabetes medicine, taking chromium dietary supplements might cause low blood sugar levels.
- Levothyroxine. Levothyroxine is a medication used to treat hypothyroidism (a condition in which the thyroid gland does not make enough thyroid hormone). Taking chromium dietary supplements together with levothyroxine might reduce the amount of levothyroxine your body absorbs so you might not be getting the full effect of the medication.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if these dietary supplements might interact with your medicines. They can also explain whether the medicines you take might interfere with how your body absorbs or uses chromium or other nutrients.

Healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about chromium?

- For more information on chromium and food sources of chromium
 - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on [Chromium](#)
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern

- [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

acne

A sometimes severe skin condition that commonly occurs on the face, neck, back, and chest and includes whiteheads, blackheads, and pimples. Severe acne may be painful and can leave permanent deep scars.

blood sugar

The main source of energy used by the body's cells. Blood sugar comes from food and is made by the liver, and is carried to the cells through the bloodstream. Also called blood glucose.

cholesterol

A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

clinical trial

A type of research study that uses volunteers to test the safety and efficacy (the ability to produce a beneficial effect) of new methods of screening (checking for disease when there are no symptoms), prevention, diagnosis, or treatment of a disease. Also called a clinical study.

control

In a clinical trial, the group of participants that does not receive the new treatment being studied. This group is compared with the group receiving the new treatment, to see whether the new treatment works. In an observational study, the controls are participants who do not have a particular health condition; the control group is compared with the group of participants who do have the condition to see if certain factors (such as diet, activity level, or use of dietary supplements) may be associated with developing or preventing the condition.

deficiency

An amount that is not enough; a shortage.

diabetes

A disease in which blood sugar (glucose) levels are high because the body is unable to use glucose properly. Diabetes occurs when the body does not make enough insulin, which helps the cells use glucose, or when the body no longer responds to insulin.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

essential

In nutrition, essential nutrients are ones that we must consume for good health because our bodies cannot make them. Essential nutrients include vitamins and minerals.

evidence

Information used to support the use of a particular screening procedure, treatment, or preventive measure. In medicine, evidence needed to determine effectiveness is provided by laboratory research, clinical trials, and other studies.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gland

A small organ that makes and releases a substance such as sweat, tears, saliva, milk, a hormone, or substances that aid in digestion.

heart attack

The blockage of an artery supplying blood and oxygen to the heart, resulting in the damage or death of a section of heart muscle.

high blood pressure

A blood pressure measurement of 140/90 mmHg (millimeters of mercury) or higher is considered high blood pressure (hypertension). Blood pressure is the force of blood pushing against the walls of the arteries. Blood pressure measurements are written as two numbers, for example 120/80. The first number (the systolic pressure) measures the pressure when the heart beats and pumps out blood into the arteries. The second number (the diastolic pressure) measures the pressure when the heart is at rest between beats. High blood pressure is a condition that occurs when a person's blood pressure often measures above 140/90 or regularly stays at that level or higher. This condition usually has no symptoms but can be life-threatening. It damages the arteries and increases the chance of stroke, heart attack, kidney failure, and blindness. Also called hypertension.

hormone

A group of chemicals made by glands in the body. Hormones circulate in the bloodstream and control the actions of certain cells or organs. Some hormones can also be manufactured.

hypothyroidism

A disorder in which the thyroid gland makes too little hormone for the body to function well. Thyroid hormones affect chemical reactions in the body, brain development, breathing, heart and nervous system functions, body temperature, muscle strength, skin dryness, menstrual cycles, cholesterol levels, and body weight.

infant

A child younger than 12 months old.

infertility

The inability to produce children.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

kidney disease

A condition that lessens the ability of the kidneys to filter wastes from the blood, keep blood chemical levels balanced, and make certain hormones. The two most common causes of kidney disease are diabetes and high blood pressure. People with long-term kidney disease may need dialysis or a kidney transplant to stay alive.

liver

A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

metabolic

Having to do with metabolism (all chemical changes that take place in a cell or organism to produce energy and basic materials needed for important life processes).

microgram

μg or mcg. A unit of weight in the metric system equal to one millionth of a gram. (A gram is approximately one-thirtieth of an ounce.)

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

ovary

One of a pair of female reproductive glands in which the eggs (ova) are formed and stored. The ovaries are located in the pelvis, one on each side of the uterus.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

prescription

A written order from a health care provider for medicine, therapy, or tests.

protein

A molecule made up of amino acids that the body needs for good health. Proteins are the basis of body structures such as skin and muscle, and substances such as enzymes and antibodies.

reproductive age

The time of life during which a person is able to conceive a child. It begins in puberty (for men and women) and ends after menopause (for women only).

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

stroke

A loss of blood flow to part of the brain. Strokes are caused by blood clots or broken blood vessels in the brain, and result in damage to a section of brain tissue. Symptoms include dizziness, numbness, weakness on one side of the body, and problems with talking or understanding language. The chance (risk) of stroke is increased by high blood pressure, older age, smoking, diabetes, high cholesterol, heart disease, a family history of stroke, and a build-up of fatty material inside the coronary arteries (atherosclerosis). See also NIH publication: Know Stroke. Know the Signs. Act in Time.

<http://www.ninds.nih.gov/disorders/stroke/knowstroke.htm>

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

thyroid

A gland located in the front of the neck, below the larynx (Adam's apple). The thyroid makes hormones that circulate in the bloodstream and affect brain development, metabolism, weight, breathing, heart rate, blood pressure, nervous system functions, body temperature, muscle strength, skin dryness, menstrual cycles, and cholesterol levels.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

Updated: March 22, 2021